1 Kernel and Injectivity

Definition 1.1 (Kernel)

The *kernel* of a linear map $T : V \to W$ is the set of vectors in V which T maps to the zero of W:

$$\ker T := \{v \in V \mid Tv = 0\} \subset V.$$

Exercise 1.2

Let $T \in \mathcal{L}(V, W)$. Show that $\ker T$ is a subspace of V.

Proof:

We have already showed $0_V \to 0_W$ under any linear map, so $0_V \in \ker T$ for T linear. Therefore, it suffices to show that $\ker T$ satisfies the subspace criterion. We have, given $v_1, v_2 \in \ker T$:

$$T(\alpha v_1 + v_2) = \alpha T v_1 + T v_2 = \alpha T(0) + T(0) = \alpha \cdot 0 + 0 = 0 + 0 = 0,$$

as desired.

Exercise 1.3

Find the kernel for each of the following linear maps.

(a) The zero map $0 : V \to W$.

(b) $T : \mathbf{C}^3 \to \mathbf{C}^3$ defined by $T(z_1, z_2, z_3) = z_1 + 2z_2 + 3z_3$. (Give a basis of $\ker T$).

(c) Let $V = \mathcal{P}(\mathbf{R})$ be the vector space of all polynomials of a real variable with real coefficients, and $D : \mathcal{P}(\mathbf{R}) \to \mathcal{P}(\mathbf{R})$ be differentiation.

(d) $T : \mathcal{P}(\mathbf{R}) \to \mathcal{P}(\mathbf{R})$ is the multiplication by x^2 map, defined by $(Tp)(x) = x^2 p(x)$.

(e) Let $T : F^\omega \to F^\omega$ be the "backward shift" map defined by $T(x_1, x_2, x_3, ...) = (x_2, x_3, ...)$.

Solution

(a) For every $v \in V$, $0v = 0$, so the kernel is $\ker(T) = V$.

(b) The kernel is the subspace defined by $\{(z_1, z_2, z_3) \mid z_1 + 2z_2 + 3z_3 = 0\}$. This is of course the equation of a plane in $\mathbf{R}^3$, with basis given by $(-2, 1, 0)$ and $(-3, 0, 1)$. From a matrix perspective, this is already an RREF matrix, and then there are two free variables and one pivot/dependent variable. So, the kernel is the subspce defined by $z_1 + 2z_2 + 3z_3 = 0$, with the above basis.

(c) These are exactly the polynomials with trivial derivative, or constants. So, the kernel is exactly the copy of $\mathbf{R}$ in the space of polynomials given by taking constants.

(d) If $x^2 p(x) = 0$, since $x^2 \neq 0$ and we have cancellation in $\mathbf{R}[x]$ (take Abstract Algebra for a proof...), we see $p(x) = 0$, so the kernel is the trivial subspace $\{0\}$.

(e) This is exactly the copy of $\mathbf{R}$ given by the sequences $(\alpha, 0, 0, 0, ...)$ for $\alpha \in \mathbf{R}$, so $\ker(T) = \mathbf{R}$.

Definition 1.4 (Injective)

A map of sets $f : A \to B$ is *injective* or *one to one* if $a_1 \neq a_2$ implies $f(a_1) \neq f(a_2)$. Or, equivalently, $f(a_1) = f(a_2)$ implies $a_1 = a_2$ (this is just the contrapositive of the first statement).

Proposition 1.5 (Injective Maps = Trivial Kernel)

A linear map $T : V \to W$ is injective if and only if $\ker T = 0$.

Exercise 1.6

Prove the above proposition.

Proof:

Injectivity is the statement that $T(v_1) = T(v_2)$ implies $v_1 = v_2$. By linearity of this map, this is equivalent to the statement that $T(v_1 - v_2) = 0$ implies $v_1 - v_2 = 0$. But, letting $v = v_1 - v_2$, this is equivalent (if and only if) to saying $T(v) = 0$ implies $v = 0$, i.e the kernel is trivial.

Exercise 1.7

Which linear map in Exercise 1.3 are injective?

Solution

The only map with trivial kernel is the multiplication by x^2 map.

2 Image and Surjectivity

Definition 2.1 (Image)

Let $T : V \to W$ be a linear map. The *image* or *range* of T is the set of all outputs of T:

$$\mathrm{Im}(T) := \{Tv \mid v \in V\} \subset W.$$

Exercise 2.2

Show that $\mathrm{Im}(T)$ is a subspace of W.

Proof:

The zero vector is an element of the image (since the zero of V maps to the zero of W). If $x, y \in \mathrm{Im}(T) \subset W$, then by definition $f(u) = x, f(v) = y$ for some $u, v \in V$. But then, we have:

$$f(\alpha u + v) = \alpha f(u) + f(v) = \alpha x + y,$$

showing $\alpha x + y \in (T)$ and therefore the subspace criterion.



Exercise 2.3

Determine the image of each of the following linear maps.

(a) The zero map $0 : V \to W$).

(b) Define $T : \mathbf{R}^2 \to \mathbf{R}^3$ by $T(x, Y) = (2x, 5y, x + y)$. Give a basis of $\mathrm{Im}(T)$.

(c) The differentiation map $D : \mathcal{P}(\mathbf{R}) \to \mathbf{P}(\mathbf{R})$.



Solution

(a) The zero map has trivial image $\{0\}$.

(b) The image has basis by $(1, 0, \frac{1}{2})$ and $(0, 1, \frac{1}{5})$.

(c) The image is exactly every polynomial (since we can integrate any polynomial), so it is $\mathbf{P}(\mathbf{R})$.



Definition 2.4 (Surjective)

A map of sets $f : A \to B$ is *surjective* or *onto* if $\mathrm{Im}(f) = B$. That is, if for every $b \in B$ there exists at least one $a \in A$ such that $b = f(a)$.



Exercise 2.5

Which of the maps in Exercise 2.3 are surjective?



Solution

The differentiation map $D : \mathcal{P}(\mathbf{R}) \to \mathcal{P}(\mathbf{R})$ is surjective.



Exercise 2.6

If we instead consider the differentiation map D on $\mathcal{P}_5(\mathbf{R})$ (the vector space of polynomials of degree at most 5), is D surjective? This shows that surjectivity of a map $T : V \to W$ depends on W.



Solution

We can never obtain a degree 5 polynomial, since if $\frac{\mathrm{d}}{\mathrm{d}x}p(x) = x^5$, then we must have $p(x) = \frac{x^6}{5} \notin \mathcal{P}_5(\mathbf{R})$